

___ Applications Development Project 3 –ADPA301
Software Requirements Specification (SRS) – Increment 1

I have read, understood and agree to the contents of this document. I accept full responsibility for any errors and omissions in this document.

Group number: _____2_____

Group name: _____Next Gen

Tech _____

	Reg no	First name	Surname	Sign	Meetings held	Number attended
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4	22332285	Akhona	Maphumulo	A.M	21	21
5	22333852	Amahle	Dunge	A. Dunge	21	21
6	22430686	Siyabonga	Mkhize	S.M	21	19
7	22435318	Anele	Madonsela	A.M	21	21
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1. Executive summary

Michaelhouse School Management System (MSMS), as we're developing it, is a centralized system designed and suited for a private boarding school, with the aim of managing its operations and commercial activities. We all know the Michaelhouse school as one of a few in South Africa that can be classified as an actual business. This system integrates various business subsystems, which include student admissions, tuition, boarding house, transport, sports, cafeteria, academic, maintenance, and events, among others. It provides a framework for different stakeholders, such as administrators, staff, parents, and students, to access their respective functions efficiently, thus enhancing its efficiency, communication, financial management, and decision-making, among other benefits, by providing a unified web-based system and mobile platform for its operations.

2. Business processes (Lists all Business processes for the project)

No	Business process	Sem / Inc	Brief description
	Student Admission Management	1/1	Parents submit the application and the admin will review and send the feedback back to the parent

	Tuition and fee billing	1/1	Parents, guardians, students, or alumni use the system to make payments for tuition, fees, and other school-related services (such as boarding, cafeteria, transport, or events).
	Academic Management	1/1	Manage all teaching and learning processes including subject allocation, attendance tracking, marks recording, result processing, and report generation.
	Order Management	1/2	Parents/students purchase essential items like school uniform, stationary, textbooks and other items needed by the student.
	Inventory Management	1/2	Manages all stock-related activities including adding new items, updating stock levels, and monitoring inventory availability to ensure smooth school operations.
	Boarding house Management	1/2	Allocate and remove students to houses and rooms with capacity control. Add and remove residences.
	Transport management system	1/2	Track vehicles, manages driver verification
	Maintenance management	2/1	Ensure the ongoing maintenance of all physical assets and building structures
	Cafeteria and meal management	2/2	Enables personalized meal services in a boarding environment that supports weekly menu planning with nutritional awareness, creating tailored plans respecting special needs and provides nutritional compliance insights.

	Sports Facility Scheduling	2/2	Book sports fields and manage facility usage.
	Event and theatre management	2/2	Manages all the bookings and organizing of internal and extended events, including the school facilities for hire.
	Alumni And Donor Financial Management	2/2	Manage alumni, donation and report financial reporting.

3. Business use cases

[Tables](#) ***Check out the Transposed Version***

No	Business process	Use case	Sem / Inc	Brief Use Case description	Full Name	Completed Yes / No
1.	<i>Student Admission Management</i>	<i>Submit Application</i>	1/1	<i>Parents will apply for their children and send the application to the admin</i>	<i>Anele</i>	Yes
2.	<i>Student Admission Management</i>	<i>Receive & Review Application</i>	1/1	<i>The admin receives the application sent by the parents and reviews it. After that the admin decides whether the applicant meets all the requirements to study at the school and sends the feedback to the applicant.</i>	<i>Azania</i>	Yes
3.	<i>Student Admission Management</i>	<i>Student Registration</i>	1/1	<i>The admin registers a student into the system after the application has been approved. The student's personal, academic, and contact details are captured</i>	<i>Anele</i>	Yes

				and stored in the system.		
4.	<i>Billing Management</i>	<i>Make Payment</i>	1/1	<i>Parents make payments for tuition fees</i>	<i>Akhona</i>	<i>Yes</i>
5.	<i>Academic Management</i>	<i>Generate Timetable</i>	1/1	<i>Admin creates timetable schedule s for classes, students and teachers</i>	<i>Lindo</i>	<i>Yes</i>
6.	<i>Academic Management</i>	<i>Record Student Attendance</i>	1/1	<i>Teacher records daily student attendance.</i>	<i>Siyabonga</i>	<i>Yes</i>
7.	<i>Academic Management</i>	<i>Capture Students Marks</i>	1/1	<i>Teacher enters marks obtained by students.</i>	<i>Mthobisi</i>	<i>Yes</i>
8.	<i>Academic Management</i>	<i>Calculates Results</i>	1/1	<i>The system calculates the student marks based on the marks captured.</i>	<i>Sihle</i>	<i>Yes</i>
9.	<i>Academic Management</i>	<i>Generate Report</i>	1/1	<i>System generates academic report.</i>	<i>Sabelo</i>	<i>Yes</i>
10	<i>Academic Management</i>	<i>Maintain Teachers Attendance</i>	1/1	<i>The system records and manages teachers' attendance to monitor staff presence during school working hours.</i>	<i>Siyabonga</i>	<i>Yes</i>
11.	<i>Order Management</i>	<i>Place order</i>	1/1	<i>Parent/student select items and place an order</i>	<i>Amahle</i>	<i>Yes</i>

12.	<i>Order Management</i>	<i>Cancel order</i>	1/2	<i>Parent/student cancels an order before it is completed</i>	<i>Azania</i>	Yes
13.	<i>Order Management</i>	<i>Track order</i>	1/2	<i>parent /student tracks the progress of an order</i>	<i>Amahle</i>	Yes
14	<i>Manage Inventory</i>	<i>Purchase Stock</i>	1/2	<i>Inventory manager purchases stock from an approved supplier to replenish inventory levels</i>	<i>Lindo</i>	Yes
15..	<i>Transport management system</i>	<i>Driver Submit Application</i>		Driver enters personal credentials and uploads required documents, then submit application.	<i>Akhona</i>	Yes
16.	<i>Transport management system</i>	<i>Receive & Review Application</i>		Admin receives driver's application, reviews the provided details and approves, rejects or requests more information.	<i>Azania</i>	Yes

17.	<i>Transport management system</i>	<i>Driver Interview</i>		<i>Transport Manager interviews a driver applicant and records the outcome (pass/fail) in the system</i>	<i>Siyabonga</i>	<i>Yes</i>
18.	<i>Transport management system</i>	<i>Plan Event</i>	<i>1/2</i>	<i>The teacher plans a school event and informs the transport team so transport can be arranged for students and staff</i>	<i>Anele</i>	<i>Yes</i>
19.	<i>Transport management system</i>	<i>Schedule Trip</i>	<i>1/2</i>	<i>The school admin or teacher schedules a student trip by entering trip details and transport requirements</i>	<i>Sihle</i>	<i>Yes</i>
20.	<i>Transport management system</i>	<i>Check List for Student</i>	<i>1/2</i>	<i>Generate digital list of students assigned to a vehicle</i>	<i>Fanele</i>	<i>Yes</i>
21.	<i>Transport management system</i>	<i>Record Driver Availability</i>	<i>1/2</i>	<i>Driver logs in to mark their availability (Available/Not Available) for a scheduled trip.</i>	<i>Mthobisi</i>	<i>Yes</i>
22.	<i>Transport management system</i>	<i>Report Vehicle Issue</i>	<i>1/2</i>	<i>Driver reports any vehicle issues found</i>	<i>Sabelo</i>	<i>Yes</i>

23.	<i>Boarding house management</i>	<i>Place Students to Residence</i>	<i>2/1</i>	<i>The system will automatically allocate students to residence and assign room numbers.</i>	<i>Mthobisi</i>	
24.	<i>Boarding house management</i>	<i>Sign in to Residence</i>	<i>2/1</i>	<i>Students scan their QR codes to record when they enter.</i>	<i>Anele</i>	
25	<i>Boarding house management</i>	<i>Sign out of Residence</i>	<i>2/1</i>	<i>Students scan their QR codes to record when they leave the residence.</i>	<i>Fanele</i>	
26	<i>Boarding house management</i>	<i>Request Visitor Access</i>	<i>2/1</i>	<i>The system allows a student to request permission for a visitor to enter the residence. The request is submitted for approval before visitor access is granted.</i>	<i>Akhona</i>	

27	<i>Boarding house management</i>	<i>Request Permission to Leave Residence</i>	<i>2/1</i>	<i>The system allows students to request permission to leave the boarding residence for a specified period.</i>	<i>Azania</i>	
28	<i>Boarding house management</i>	<i>Alert Students Of Emergency</i>	<i>2/1</i>	<i>The system notifies all students of an emergency by sending an emergency alert to the residence application. The alert instructs students to proceed to the designated emergency assembly point</i>	<i>Lindo</i>	
29	<i>Boarding house management</i>	<i>Confirm Emergency Safety</i>	<i>2/1</i>	<i>students mark themselves as safe during an emergency. The system verifies their location at the assembly point and provides House Masters with real-time accountability</i>	<i>Siyabonga</i>	

30	<i>maintenance management</i>	<i>Report Asset Failure</i>	<i>2/1</i>	<i>A user will be able to digitally log a fault when an asset (e.g., a light bulb or Pipe Burst) fails.</i>	<i>Sihle</i>	
31	<i>maintenance management</i>	<i>Auto-Assign Maintenance Staff</i>	<i>2/1</i>	<i>Based on the failure type and staff availability, the system triggers alerts to the most suitable staff.</i>	<i>Akhona</i>	
32	<i>maintenance management</i>	<i>Complete Job Card</i>	<i>2/1</i>	<i>The assigned staff member receives the notification, performs the repair, and updates the status to complete within the app</i>	<i>Sabelo</i>	
33	<i>maintenance management</i>	<i>Schedule Recurring Maintenance</i>	<i>2/1</i>	<i>The system monitors time intervals (e.g., 6 months for AC filters, 2 week for Grass cutting) and automatically creates a new job card without human intervention.</i>		

34	<i>Cafeteria and meal management</i>	<i>Schedule Weekly Menus</i>	<i>2/2</i>	<i>Cafeteria manager plans and publishes weekly menus while considering ingredient availability and nutritional balance.</i>	<i>Anele</i>	
35	<i>Cafeteria and meal management</i>	<i>Create Personalized Meal Plan</i>	<i>2/2</i>	<i>Students create meal plans based on dietary preferences, allergies, medical conditions and sports requirements.</i>	<i>Sabelo</i>	
36	<i>Cafeteria and meal management</i>	<i>Verify Meal Collection</i>	<i>2/2</i>	<i>When students or staff collect meals, the system verifies their order using a facial recognition, QR code to ensure the correct meal is issued.</i>	<i>Sabelo</i>	

37	<i>Cafeteria and meal management</i>	<i>Assess Nutritional Compliance</i>	2/2	<i>The system monitors whether students' selected meals meet their daily nutritional requirements and alerts cafeteria staff when adjustments are required for students with special dietary needs.</i>	<i>Lindo</i>	
38	<i>Cafeteria and meal Management</i>	<i>Verify & Approve Meal Plan</i>	2/2	<i>This system permits the Cafeteria Manager or Dietitian to check the meal plan produced to confirm that it meets the dietary preferences, allergies, health problems, and availability of the ingredients among others before it is accepted.</i>	<i>Anele</i>	

39	<i>Cafeteria and meal Management</i>	<i>Generate Feast Plan</i>	2/2	<i>Created from the combination of all the approved meal plans and other important factors such as the number of guests, dietary preferences, and nutritional values. The feast plan helps the cafeteria prepare the appropriate amount and range of meals.</i>	<i>Lindo</i>	
40	<i>Sports & Facility Scheduling</i>	<i>Book Facility (Internal)</i>	2/2	<i>Sport Staff reserve a sports field or indoor centre for school practices or matches.</i>	<i>Akhona</i>	
41	<i>Sports & Facility Scheduling</i>	<i>Request Facility Hire (External)</i>	2/2	<i>An external organization (e.g., local club) requests to rent a school facility for a private event.</i>	<i>Fanele</i>	
42.	<i>Sports & Facility Scheduling</i>	<i>Issue Digital Access Permit</i>	2/2	<i>System generates a time-bound QR entry code for external users once payment is verified</i>	<i>Mthobisi</i>	

43.	<i>Events & Theatre Management</i>	<i>Supports event scheduling</i>	<i>2/2</i>	<i>Personnel schedules events according to facility availability</i>	<i>Sihle</i>	
44	<i>Events & Theatre Management</i>	<i>Supports facility booking</i>	<i>2/2</i>	<i>User books a facility for an event according to slot availability</i>	<i>Lindo</i>	
45.	<i>Events & Theatre Management</i>	<i>Allocate Seating based on capacity</i>	<i>2/2</i>	<i>The admin approves the number of seats allocated that can carry the event capacity</i>	<i>Sabelo</i>	
46	<i>Events & Theatre Management</i>	<i>Generate revenue reports</i>	<i>2/2</i>	<i>The user has the option to report the revenue generated from the event to support potential future sponsorship</i>	<i>Fanele</i>	
47	<i>Alumni & Donor Financial Management</i>	<i>Register Alumni</i>	<i>2/2</i>	<i>Capture and store alumni personal and contact information in the system.</i>	<i>Anele</i>	
48.	<i>Alumni & Donor Financial Management</i>	<i>Create Fundraising Campaign</i>	<i>2/2</i>	<i>Set up a new fundraising campaign with goals, dates, and target audience.</i>	<i>[1] Azania</i>	
49.	<i>Alumni & Donor Financial Management</i>	<i>Record Donation</i>	<i>2/2</i>	<i>Capture and store details of a donation made by a donor.</i>	<i>[2] Sihle</i>	

50	<i>Alumni & Donor Financial Management</i>	<i>Create Pledge</i>	<i>2/2</i>	<i>Record a donor's promise to contribute a specific amount in the future.</i>	<i>[3] Akhona</i>	
51	<i>Alumni & Donor Financial Management</i>	<i>Convert Pledge</i>	<i>2/2</i>	<i>Update a pledge status when the promised amount is paid.</i>	<i>[4] Amahle</i>	
52.	<i>Alumni & Donor Financial Management</i>	<i>Analyze Campaign Statistics</i>	<i>2/2</i>	<i>Analyse campaign performance metrics such as funds raised and donor activity.</i>	<i>[5] Lindo</i>	
53.	<i>Alumni & Donor Financial Management</i>	<i>Generate Donor Score</i>	<i>2/2</i>	<i>Calculate and assign a score to donors based on giving history and engagement.</i>	<i>Siyabonga</i>	
54.	<i>Alumni & Donor Financial Management</i>	<i>Generate ROI Report</i>	<i>2/2</i>	<i>Produce a report showing the return on investment for fundraising campaigns.</i>	<i>Mthobisi</i>	
55.	<i>Alumni & Donor Financial Management</i>	<i>Export Audit Report</i>	<i>2/2</i>	<i>Generate and download a detailed report for financial review and compliance purposes.</i>	<i>Sabelo</i>	

4. Use case description (**Only describing Semester 2 Increment 1 use cases**)

Use case number	23
Use case name	<i>Place Students to Residence</i>
Brief description	<i>The system automatically allocates students to residences immediately after successful registration.</i>
Triggering event	<i>Student becomes successfully registered</i>
Actor(s)	<i>Residence management system</i>
Preconditions	1. <i>Students must be registered in the system.</i> 2. <i>Students have not been allocated to the system.</i> 3. <i>Bed must be available</i>
Flow of activities	1. <i>The student completes registration.</i> 2. <i>The system creates student account.</i> 3. <i>The residence management system is triggered automatically, immediately after registration.</i> 4. <i>The system searches for residences with vacant beds</i> 5. <i>The system automatically selects the most suitable residence according to allocation rules.</i> 6. <i>The system assigns residence and room.</i> 7 <i>The system updates residence, room occupancy and bed availability</i> 8 <i>The registration process completes.</i>

Use case number	24
Use case name	Sign In To Residence
Brief description	Students scan their QR codes to record when they enter
Triggering event	Student scans their QR code at the residence entrance or exit.
Actor(s)	Student
Preconditions	1. Student must be registered in the system. 2. Student must have a valid QR code. 3. QR code scanner must be available.
Flow of activities	1. Student arrives at the residence. 2. Student scans their QR code. 3. System verifies the QR code. 4. System records the student's sign-in . 5. System displays a confirmation message.

Use case number	25
Use case name	Sign Out Of Residence
Brief description	Students scan their QR codes to record when they leave the residence.
Triggering event	Student scans their QR code at the residence exit.
Actor(s)	Student
Preconditions	1. Student must be registered in the system. 2. Student must have a valid QR code. 3. QR code scanner must be available.

Flow of activities	1. Student leaves the residence. 2. Student scans their QR code. 3. System verifies the QR code. 4. System records the student's sign-out time. 5. System displays a confirmation message.
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Use case number	26
Use case name	<i>Request Visitor Access</i>
Brief description	The system allows a student to request permission for a visitor to enter the boarding residence. The student provides the required visitor and visit details, after which the system validates the request and submits it for approval.
Triggering event	A student wants to have a visitor enter the boarding residence.
Actor(s)	Student Housemaster
Preconditions	

Flow of activities	1. Student logs into the residence application. 2. Student selects Request Visitor Access. 3. Student enters the visitor's details. 4. Student specifies the intended date and time of the visit. 5. Student provides the reason for the visit. 6. Student submits the visitor access request. 7. System validates the submitted visitor and visit information. 8. System checks whether the requested visit complies with the residence's visitor rules and permitted visiting times. 9. System creates the visitor access request with a Pending status. 10. System sends the request to the House Master for review. 11. House Master reviews the visitor access request. 12. House Master approves or rejects the request. 13. System records the decision. 14. System notifies the student of the outcome. 15. If approved, the system makes the visitor eligible for the temporary visitor access process.
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Use case number	27
Use case name	<i>Request Permission to Leave Residence</i>
Brief description	<i>The system allows students to request permission to leave the boarding residence for a specified period.</i>
Triggering event	

Actor(s)	Student, Parents And Housemaster
Preconditions	
Flow of activities	

Use case number	28
Use case name	Confirm Emergency Safety
Brief description	The system enables students to confirm their safety during an emergency by marking themselves as safe when they are within the designated emergency assembly area. The system uses geofencing to verify the student's presence at the assembly point and provides the House Master with an emergency accountability overview.

Triggering event	An emergency is declared and the emergency safety function is activated.
Actor(s)	Student House Master
Preconditions	1. Student must be registered in the system. 2. Student must be logged into the residence application. 3. The House Master must have access to the emergency safety function. 4. A geofence must be configured around the designated emergency assembly area. 5. The emergency safety function must be active.
Flow of activities	1. House Master activates the emergency safety function. 2. System activates the geofence around the designated emergency assembly area. 3. Student opens the emergency safety function. 4. Student selects Mark Me as Safe. 5. System checks whether the student is within the designated emergency geofence. 6. If the student is within the geofence, the system records the student as Safe and Accounted For. 7. System records the date and time of the safety confirmation. 8. System updates the student's emergency status. 9. System compares the students who have confirmed their safety with the residence student records. 10. System displays the number of students who have confirmed their safety and the students who have not yet responded. 11. House Master reviews the emergency accountability overview. 12. System stores the emergency safety confirmation record.

Use Case number	29
Use case name	Schedule Weekly Menus
Brief description	The Cafeteria Manager, with help from the Dietitian and Kitchen Staff, prepares and publishes the weekly menu. The menu is based on student menu drafts, healthy meal guidelines, available ingredients, and the number of meals expected.
Triggering event	The weekly menu needs to be finalised after students submit their menu drafts, usually on Tuesday afternoon.
Actor(s)	Cafeteria Manager, Dietitian, Kitchen Staff, System
Preconditions	· Students have submitted their menu drafts through the Prefects. · The meal catalogue, nutrition information, and supplier inventory are available. · The Student Profiles, Sports, and Boarding House systems are connected.

Flow of activities	1. Cafeteria Manager logs into the system. 2. Selects "Schedule Weekly Menus". 3. System displays suggested menus based on historical data, upcoming sports/events (from Sports & Events subsystems), and projected demand. 4. Manager chooses or modifies meals for each day, applying templates (e.g., standard, vegetarian, high-protein). 5. System checks real-time ingredient availability (Inventory subsystem) and flags shortages. 6. System validates nutritional balance and highlights conflicts with common dietary needs. 7. Manager reviews projected demand (aggregated from student registrations, boarding houses, and pre-orders). 8. Cafeteria Manager confirms and publishes the menu to students, staff, parents, and House Parents via portal/app. 9. System stores the schedule and pushes data to Personalised Meal Plan generation and Inventory forecasting.
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Use Case number	30
Use case name	Create Personalised Meal Plan
Brief description	The system automatically generates (and allows manual adjustment of) personalised meal plans for students based on their menu drafts, profile, dietary preferences, allergies, medical conditions, sports requirements, and boarding house data, with dietician and parent/guardian visibility and consent.

Triggering event	Student registration/enrolment update, start of term, sports season change, or manual request by authorised staff/parent.
Actor(s)	Student, Dietitian, Cafeteria Manager, Kitchen Staff, Parent/Guardian, System.
Preconditions	1. Student is registered in the Student Registration subsystem with up-to-date profile. 2. Student dietary, medical, and sports profile exists 3. Meal catalogue and weekly menu are available. 4. Nutritional information and allergen data for meals are stored.
Flow of activities	1. Authorised user (Cafeteria Manager, Dietician, or Parent) logs in and selects "Generate Personalised Meal Plan". 2. System retrieves student data: dietary preferences, allergies, medical conditions, sports participation, and boarding house. 3. System analyses available meals from the published weekly menu and generates plan using templates and rules. 4. System flags conflicts and suggests alternatives. 5. Cafeteria Manager / Dietician reviews and approves. 6. System saves the meal plan, makes it visible to the student and House Parent, and feeds data into demand forecasting and nutritional compliance. 7. System sends notification to student/parent/House Parent

Use Case number	31
Use case name	Verify Meal Collection

Brief description	<i>The cafeteria team must make sure that the student's approved meal plan (or menu) corresponds to the meal served, keeping track of inventory and consumption, and updating the records of suppliers and clients if necessary. A student or staff member needs to come for the meal.</i>
Triggering event	<i>A student or staff member arrives to collect a meal or pre-ordered item.</i>
Actor(s)	<i>Cafeteria Staff, Student, House Parent/Guardian, System</i>
Preconditions	<i>1. The student or staff member must be registered.</i> <i>2. The meal allocation or pre-order must be present.</i> <i>3. Student card, QR code, or biometric identification methods should be available.</i>
Flow of activities	<i>1. The cafeteria team must invoke "Meal Collection Verification."</i> <i>2. The student must present his ID or QR code.</i> <i>3. The system must show the approved meal plan of the student and any special instructions.</i> <i>4. The staff must verify and serve the food.</i> <i>5. The system must track the consumption of the meal, update inventory, and record the food used for nutritional purposes.</i> <i>6. The system must update daily demand information for the supplier. Quick feedback on the meal can be obtained.</i> <i>7. System logs any substitutions or no-shows for waste analysis and future forecasting.</i> <i>8. Optional: Student rates the meal via quick feedback on the app.</i>

Use case number	32
Use case name	<i>Verify & Approve Meal Plan</i>
Brief description	<i>The system allows the Cafeteria Manager or Dietitian to review generated meal plans to ensure they meet students' dietary preferences, allergies, medical conditions, nutritional requirements, and ingredient availability before they are approved for use.</i>
Triggering event	<i>A personalised meal plan is generated or an existing meal plan is updated and requires approval.</i>
Actor(s)	<i>Cafeteria Manager, Dietician, House Parent, System</i>
Preconditions	<i>1. A personalised meal plan must have been generated.</i> <i>2. The student must be registered in the system.</i> <i>3. The student's dietary preferences, allergies, medical conditions, and nutritional requirements must be available.</i> <i>4. The weekly menu and ingredient inventory must be available.</i>
Flow activities of	<i>1. The Cafeteria Manager or Dietitian logs into the system.</i> <i>2. The user selects Verify & Approve Meal Plan.</i> <i>3. The system retrieves the generated meal plan and the student's dietary profile.</i> <i>4. The system validates the meal plan against allergies, medical conditions, dietary preferences, sports requirements, and nutritional guidelines.</i> <i>5. The system checks ingredient availability and identifies any unavailable items.</i> <i>6. The system highlights any conflicts or recommendations requiring attention.</i> <i>7. The Cafeteria Manager or Dietitian reviews the meal plan and recommendations.</i> <i>8. The user approves the meal plan or modifies it before approval.</i> <i>9. The system updates the meal plan status to Approved.</i> <i>10. The system makes the approved meal plan available to the kitchen staff, meal collection process, and the student or parent through the portal.</i> <i>11. The system stores the approval record, including the approver's details, date, and time.</i>

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Use Case number	33
Use case name	<i>Assess Nutritional Compliance</i>
Brief description	<i>The system confirms that the right meal is distributed to the right student or staff member, tracks the inventory, and logs the distribution into the system for future reference.</i>
Triggering event	<i>Creating or updating meals, collecting meals at the end of a day, week, or reviewing them manually.</i>
Actor(s)	<i>Cafeteria Manager, Dietitian, Event Coordinator, System</i>
Preconditions	<i>1. Receivable student nutrition profile should be present</i> <i>2. Data on meal plan and collection should be available</i> <i>3. Data on nutritional values should be recorded in the system</i>

Flow of activities	1. Staff authorized by the manager or the system manually finding out about nutritional compliance. 2. The system retrieves student profile, meal and collection information. 3. Checks whether meals offered are aligned with the meals required 4. Determining what is lacking and needs to be followed through (e.g. the risk of allergy, low protein for the sport players, etc.) 5. The system sends out alerts and recommendations 6. Staff being informed about the results changes the nutrition plan 7. System updates compliance status and keeps information on further reports. 8. System flags students needing follow-up for long-term monitoring.
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Use Case number	33
Use case name	Generate Feast Plan
Brief description	The system generates a comprehensive feast plan for school events and special occasions by consolidating approved meal plans, guest attendance, dietary requirements, nutritional needs, and available inventory. The feast plan assists the cafeteria team in preparing the correct quantities and variety of meals.
Triggering event	An event requiring catering is approved, or the Cafeteria Manager requests the generation of a feast plan.
Actor(s)	Cafeteria Manager, Dietitian, System

Preconditions	1. The event must be approved and scheduled in the system. 2. Approved meal plans must be available for attendees. 3. Event attendance records must be available. 4. Inventory and ingredient information must be up to date. 5. Nutritional and dietary information must exist for attendees.
Flow activities of	1. The Cafeteria Manager selects Generate Feast Plan. 2. The system retrieves the selected event details and expected attendance. 3. The system gathers all approved meal plans for students, staff, guests, and VIP attendees. 4. The system analyses dietary preferences, allergies, medical conditions, and special nutritional requirements. 5. The system checks ingredient availability and current inventory levels. 6. The system calculates the required quantities of each meal, ingredient, and beverage. 7. The system generates an optimized feast plan, including menu items, preparation quantities, serving schedule, and staffing recommendations. 8. The Cafeteria Manager reviews the generated feast plan. 9. The Cafeteria Manager approves the feast plan or makes manual adjustments if necessary. 10. The system stores the approved feast plan and makes it available to the kitchen staff, inventory management, and event management subsystems for execution.

Use Case number	34
Use case name	Schedule Recurring Maintenance
Brief description	The system automatically monitors predefined time intervals and generates a new job card for routine maintenance tasks without requiring human intervention.

Triggering event	The system clock reaches a predefined maintenance interval (e.g., 6 months since last AC filter change or 2 weeks since last grass cutting).
Actor(s)	System (Scheduler)
Preconditions	1. The system is configured with specific time intervals (e.g., 6 months for AC filters, 2 weeks for Grass cutting). 2. Assets and maintenance templates are registered in the system
Flow of activities	1. The system continuously monitors the internal clock and maintenance schedules. 2. The system detects that a defined time interval has elapsed since the last maintenance. 3. The system automatically creates a new job card for the specific asset. 4. The system populates the job card with relevant details (asset ID, maintenance type, due date). 5. The system stores the job card and triggers the assignment process.

Use Case number	35
Use case name	Report Asset Failure
Brief description	This use case allows any authorized user to digitally log a fault. The user provides details about the failure, allowing the system to create a new issue ticket.
Triggering event	A user discovers a faulty asset (e.g., a burst pipe or broken light bulb) and initiates the reporting process.

Actor(s)	User (Primary): Reports the failure. Maintenance Manager (Secondary): Receives the report.
Preconditions	1. The user is authenticated in the system. 2. The asset exists in the asset database.
Flow of activities	1. The user navigates to the "Report Fault" section of the app. 2. The user selects the specific asset or scans its QR code. 3. The user enters a description of the fault (e.g., "Water leaking," "No power"). 4. The user selects the severity level (e.g., Low, Medium, High). 5. The user submits the report. 6. The system creates a new "Failure Report" and assigns it a unique ID. 7. The system notifies the Maintenance Manager. Alternative Flow (Asset Not Found): If the user cannot find the asset in the list, they can add a manual entry with a general location and asset name.

Use Case number	36
Use case name	Auto-Assign Maintenance Staff
Brief description	Based on the failure type and current staff availability, the system triggers alerts and automatically assigns the most suitable staff member to the job.
Triggering event	A new failure report or job card is created and requires staffing.

Actor(s)	System (Primary): Performs the assignment logic. Maintenance Staff (Secondary): Receives the assignment alert.
Preconditions	1. A job card with a specific "Failure Type" exists. 2. Staff member profiles include their skills/expertise and current workload.
Flow of activities	Basic Flow: 1. The system analyzes the "Failure Type" (e.g., Electrical, Plumbing). 2. The system queries the database for staff with matching skills. 3. The system filters the list to only those who are currently "Available" (not on leave or overloaded). 4. The system assigns the job to the most suitable staff member. 5. The system sends a push notification to the assigned staff member with the job details. Alternative Flow (No Suitable Staff): If no staff are available with the required skills, the system notifies the Maintenance Manager to manually assign the task or outsource it.

Use Case number	37
Use case name	Complete Job Card
Brief description	After a fault is reported and system assigns it to a maintenance worker, the worker receives a notification, then they go to the location, fix the problem, take a photo before and after the repair, write what they did, and tap Complete. The job card then closes automatically and the person who reported the fault is notified that it has been fixed.

Triggering event	A maintenance worker receives a notification on their phone telling them they have been assigned a new hob.
Actor(s)	Maintenance worker, System
Preconditions	• A job card must have been created in the system • The job card must have been assigned to the specific worker • The worker must be registered in the system and logged in

Flow of activities	1. The worker receives a push notification saying they a new job assigned to them. 2. The worker opens the system and taps on the job to see the full details. 3. The system shows the following information: • What the problem is • Exactly where to go in on the school • What priority level it is (Emergency, High, Medium, or Low) • Who reported it and when 4. The worker goes to the location of the problem. 5. Before starting the repair, the worker takes a photo of the fault using their phone camera inside the app. This is the before photo showing a proof of what is broken. 6. The worker fixes the problem 7. After fixing it, the worker takes another photo showing the repair is done. This is the after-photo showing proof of that the job was completed properly. 8. The worker types a short completion note explaining what they did and what parts they used if any. 9. The worker selects the final condition of the asset from list: 2 • Fixed - problem is fully resolved • Repaired Temporarily - a short-term fix was done, needs a permanent fix later • Needs Further Work - the problem is bigger than expected, more help is needed • Replaced - the broken part was removed and replaced with a new one 10. The worker taps Mark as Complete 11. The system records the exact time the job was completed. 12. The system calculates how long the job took from when it was reported to when it was completed. 13. The system updates the asset history to show this repair was done. 14. The system sends a notification to the person who originally reported the fault, telling them the job has been fixed. 15. The job card status changes to Completed and moves to the archive.
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Use Case number	38
Use case name	Track Project Performance
Brief description	This use case utilizes a dashboard to monitor the progress, costs, and efficiency of various maintenance projects, allowing managers to make data-driven decisions.
Triggering event	A user (Manager) requests to view the current state of maintenance projects.
Actor(s)	Maintenance Manager / Admin (Primary): Views the dashboard.
Preconditions	1. The user has administrative or managerial permissions. 2. Historical and current Job Card data exists in the system.
Flow of activities	Basic Flow: 1. The user navigates to the "Dashboard" or "Performance" module. 2. The user selects a date range or specific project. 3. The system loads and displays key metrics (KPI dashboards). 4. The data includes: Total Jobs Completed, Average Response Time, Jobs Pending, and Maintenance Costs. 5. The user can filter the view to see data for specific trades or staff. Alternative Flow (Data Export): The user can click an "Export" button to generate a PDF/Excel report for meeting presentations.

Use Case number	39
Use case name	Manage Preventive Schedules
Brief description	This use case digitizes traditional maintenance diaries. Using a Time Planner tool, managers can schedule inspections and services, ensuring that compliance dates and deadlines are never missed.
Triggering event	A Maintenance Manager needs to plan upcoming inspections or services to ensure compliance.
Actor(s)	Maintenance Manager (Primary): Creates and manages schedules. System (Secondary): Displays the planner and sends reminders.
Preconditions	1. The user has permission to create schedules. 2. Regulatory or internal compliance dates are known.
Flow of activities	Basic Flow: 1. The manager navigates to the "Time Planner" / "Schedule" module. 2. The manager selects a specific date/time on the calendar. 3. The manager enters the task details (e.g., "Fire Extinguisher Inspection - Building A"). 4. The manager sets the recurrence pattern (e.g., Monthly, Quarterly) if applicable. 5. The system saves the schedule and logs the compliance requirement. 6. The system automatically schedules a reminder (email/push) for 24 hours before the task is due. Alternative Flow (Compliance Alert): If a compliance deadline is missed (not marked done by the due date), the system highlights the task in red on the dashboard and escalates an alert to senior management.

UseCase number	40
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UseCase name	Book Facility
Brief description	Sport staff reserve a sports field or indoor centre for school practices or matches.
Triggering event	A coach initiates a booking request.
Actor(s)	Coach, Student
Preconditions	The facility(Field or indoor court) must exist in the system.
Flow of activities	1. Staff clicks "New Booking" and selects the facility type (Field or Indoor Centre). 2. Staff inputs the desired Date, Start Time, and End Time. 3. The system immediately refreshes the calendar view, greying out unavailable slots and highlighting only available time windows in green. 4. Staff selects the specific Sport (e.g., Rugby, U16 Basketball). 5. Staff clicks "Confirm Reservation." 6. System checks for double-booking one final time, locks the slot, and sets status to "Confirmed" (no approval workflow required). 7. The system sends a confirmation email to the coach.

UseCase number	41
UseCase name	Verify Facility Readiness
Brief description	The system verifies that a sports facility is ready for use by checking its maintenance status, equipment availability, safety requirements, and booking schedule before the activity begins.
Triggering event	A scheduled sports activity is approaching and the facility readiness verification process is initiated.
Actor(s)	Sports Staff, Maintenance Manager, System

Preconditions	1.The sports facility must be registered in the system. 2.A facility booking must exist. 3.Maintenance records must be available. 4.Required sports equipment must be allocated.
Flow of activities	1.The Sports Staff selects Verify Facility Readiness. 2.The system retrieves the facility booking details. 3.The system checks whether the facility has any outstanding maintenance issues. 4.The system verifies that the required sports equipment has been allocated. 5.The system checks that the facility is available and free from scheduling conflicts. 6.The system performs a safety readiness check using the latest inspection records. 7.If all requirements are satisfied, the system marks the facility as 8.Ready for Use. 9.If any issues are detected, the system generates an alert and notifies the Sports Staff and Maintenance Manager. 10.The system records the readiness verification date and time. 11.The system makes the facility status available to coaches and event coordinators.

UseCase number	42
UseCase name	Book Sports Equipment
Brief description	The system allows coaches or sports staff to reserve sports equipment required for training sessions, matches, or school events while ensuring equipment availability and preventing double-booking.
Triggering event	A coach or sports staff member requires equipment for a scheduled practice, match, or event.
Actor(s)	Coach, Sports Staff, System

Preconditions	1.The coach or sports staff member must be registered in the system. 2 Sports equipment must be registered in the inventory. 3.The equipment must be available for the requested date and time. 4.The related sports facility booking must exist or be created.
Flow of activities	1.The coach or sports staff member must be registered in the system. 2. Sports equipment must be registered in the inventory. 3. The equipment must be available for the requested date and time. 4. The related sports facility booking must exist or be created. 5.The coach selects the booking date and time. 6.The system checks equipment availability. 7.If the equipment is available, the system reserves the equipment. 8.The system links the equipment booking to the scheduled sports activity. 9.The system updates the equipment availability status. 10. The system sends a booking confirmation to the coach.

Use case number	42
Use case name	Issue Digital Access Permit
Brief description	The system automatically generates a secure, time-limited QR code that will serve as a digital access permit for entry into school premises after confirming that the external user's facility booking has been approved and payment has been verified.

Triggering event	Successful payment of the booking fee for the facility.
Actor(s)	System
Preconditions	1. The facility booking has been approved 2. Payment has been verified successfully 3. The booking date and time must be confirmed
Flow of activities	1. The system detects that payment has been verified 2. The system retrieves the approved booking details 3. The system generates a unique digital access permit 4. The system generates a secure QR code 5. The QR code is sent to the external user via email

UseCase number	43
UseCase name	Supports Event Scheduling
Brief description	Personnel schedules events according to facility availability. The system displays available time slots and prevents double-booking, ensuring that events are planned with conflicts
Triggering event	A user(event coordinator/Admin) needs to schedule a new event and select a facility or venue
Actor(s)	Event Coordinator, Admin, System

Preconditions	1. The user is logged in with appropriate permissions. 2. Facilities and venues are registered in the system. 3. Event Details(name, date, time, expected attendance, purpose) are available
Flow of activities	1. The user navigates to the Event Scheduling module. 2. The user selects a facility or venue from the list. 3. Systems displays time slots based on existing booking and facility capacity 4. User selects a preferred time slot 5. Users enter event details (name, date,time, description, expected attendees, special requirements). 6. The user submits the scheduling request. 7. System validates that the selected slot is still available 8. The system reserves the slot and creates a new event record. 9. The system updates the facility calendar. 10. System sends a confirmation notification to the user and relevant stakeholders 11. The event appears on the institutional calendar and facility schedule

UseCase number	44
UseCase name	Supports Facility Booking
Brief description	
Triggering event	
Actor(s)	
Preconditions	
Flow of activities	

UseCase number	45
UseCase name	Allocate Seating based on capacity
Brief description	
Triggering event	
Actor(s)	
Preconditions	
Flow of activities	

UseCase number	46
UseCase name	Verify Attendant
Brief description	The system verifies the identity of event attendants using facial recognition technology before granting access to the event. The system confirms that the attendee is registered for the event, records their attendance, and prevents unauthorized or duplicate entry.
Triggering event	An attendee arrives at the event entrance and initiates the verification process by presenting themselves for facial scanning.
Actor(s)	Attendant, Event Staff, System
Preconditions	1.The event must be scheduled and active. 2.The attendant must be registered for the event. 3.The attendant's facial biometric profile must be stored in the system. 4.A facial recognition device or camera must be available and connected to the system.

Flow of activities	1.The attendant arrives at the event entrance. 2.The Event Staff selects Verify Attendant. 3.The system activates the facial recognition scanner. 4.The attendant faces the camera for biometric verification. 5.The system compares the captured facial image with the registered biometric profile. 6.The system verifies that the attendant is registered for the selected event. 7.The system checks whether the attendant has already been admitted to the event. 8.If the verification is successful, the system records the date and time of entry. 9.The system updates the attendant's attendance status to .Verified or Checked In. 10.The system displays a confirmation message and grants access to the event. 11.If verification fails, the system denies access and notifies the Event Staff to perform manual verification. 12.The system stores the verification record for attendance reporting and event analytics.
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UseCase number	46
UseCase name	Generate Revenue Report
Brief description	
Triggering event	
Actor(s)	
Preconditions	

Flow of activities	
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Use Case number	47
Use case name	Generate Donor Score
Brief description	The System calculates and assigns a score to each donor based on their giving history, engagement level, and interaction with the institution. This score helps the institution identify high-value donors and tailor fundraising strategies accordingly
Triggering event	A fundraising or alumni officer requests a donor score report for a specific donor or a group of donors.
Actor(s)	Fundraising Officer, Alumni Officer, System
Preconditions	1. The user is logged in with appropriate permissions. 2. Donor records exist in the system. 3. Donation and engagement history data is available and up-to-date.

Flow of activities	1. User navigates to Donor Management module and selects Generate Donor Score. 2. User specifies the donor(s) (single donor, a group, or all donors). 3. User optionally filters by date range, donation amount, or engagement type. 4. User clicks Generate Score. 5. System retrieves the donor's giving history (frequency, total amount, largest donation, recency). 6. System retrieves engagement data (event attendance, communication, volunteer activity). 7. System calculates the donor score using a weighted algorithm (e.g., 60% giving history, 30% engagement, 10% recency). 8. System assigns a score (e.g., 0–100) to the donor(s). 9. System displays the donor score(s) in a dashboard view, with an optional breakdown of the scoring components. 10. User can export the scores as a report (PDF/Excel) or use the data to plan targeted fundraising campaigns.
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UseCase Number	47
UseCase Name	
Brief Description	
Triggering Event	
Actor(s)	
Preconditions	
Flow of activities	

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Use case number	48
Use case name	Generate RIO Report
Brief description	The fundraising manager requests a Return on Investment report. The system analyses fundraising campaign costs and donations received, calculates RIO, and generates a report that helps management evaluate campaign performance.
Triggering event	The fundraiser manager selects Generate RIO Report
Actor(s)	Fundraising manager.
Preconditions	1. Campaign information exists 2. Donation records have been captured 3. campaign expenses have been recorded
Flow of activities	1. The fundraising manager selects Generate RIO report 2. The system displays available fundraising campaigns 3. The manager selects a campaign 4. The system retrieves campaign costs and donation records 5. The system calculates the campaign's RIO. 6 The system displays the report 7 The manager downloads or prints the report

Please Click link Below to Access the minutes

Meeting Minutes

SEMESTER 2 Meeting Minutes

Meeting Minutes for Increment 1

Meeting 1

1. Date and time of the meeting: 16/07/2026 ,20:00 pm.
2. Duration of the meeting: 1hr 30min
3. Attendance –(Present / Absent): All present
4. Venu: Microsoft teams
5. Minutes: -Discussion of the individual submission of the Use Cases that will be due in the first week.
-The Boarding House Management and Maintenance management business descriptions are all clear to the group members they don't need to be refined.
Cafeteria Meal plan business process is not clear enough to the group so it needs to be refined and that will be done by Lindokuhle and he will explain it on the next meeting.
- 6.Meeting closes; 21:30 pm.

Meeting Minutes for Increment 1

Meeting 2

1. Date and time of the meeting: 20/07/2026 ,19:30 pm.
2. Duration of the meeting: 1hr 30min
3. Attendance –(Present / Absent): All present
- 4.Venu: Microsoft teams
5. Minutes: - Discussion of use cases for increment 1.
- Assigning Use Cases to group members .
-Lindokuhle explaining Cafeteria Management business process and he introduces these Use Cases, Create Personalised Meal Plan, Verify Meal Collection, Schedule Weekly Menus, Monitor Nutritional Requirements.
- Meal plan schedule will change in every week in order to accommodate all students.
-The system will automatically detect the meals plan for students with a strict diet e.g health and sports students.
- 6.Meeting closes; 20:30 pm.

Meeting Minutes for Increment 1

Meeting 3

1. Date and time of the meeting: 31/07/2026 ,20:30 pm.
2. Duration of the meeting: 30min
3. Attendance –(Present / Absent): Siyabonga Mkhize -Absent
4. Venu: Microsoft teams
5. Minutes:- Cafeteria Management business process will not be done in the first increment of this semester.
 - The reason why Cafeteria Management business process is out it is because its Use Cases were not approved by the lectures.
 - Group members that were assigned to Cafeteria Use Cases will now join the remaining Business processes which are Boarding House Management and Maintenance Management.
 - Members have to start coding and by Tuesday the code must be represented in the meeting that will be held.
- 6.Meeting closes; 21:00 pm.

Meeting Minutes for Increment 1

Meeting 4

1. Date and time of the meeting: 04/08/2026 ,20:30 pm.
2. Duration of the meeting: 1hr 40min
3. Attendance –(Present / Absent): Siyabonga Mkhize -Absent
4. Venu: Microsoft teams
5. Minutes:- The meeting is about discussing the final Use Cases for increment 1 and also showing and reviewing how far the members are with the coding as it was mentioned in the previous meeting.
 - The Student Allocation Use Case that falls under the Boarding House Management business process will be splitted into two Use Cases because it is a complicated Use Case and it requires a lot of coding.
 - Visitor Management Use Case has been introduced as a new use case that will be under the Boarding House Management business

process.

- On the coding side Mthobisi has already pushed the code for Student Allocation Use Case but the code to the main branch of the group on visual studio doesn't work at the moment because it have some errors that needs to be fixed.
- .- The Maintenance management business process Sihle has already coded its Use Cases and they are all functioning but he couldn't be able to show it into the meeting due to bad internet connection.
- The group still needs two more Use Cases that will fall under Boarding House Management for incement 1 so all members have been assigned to search for them.

6.Meeting closes; 21:40 pm.

Meeting Minutes for Increment 1

Meeting 5

1. Date and time of the meeting: 06/08/2026 ,10:00 am.
2. Duration of the meeting: 1hr
3. Attendance –(Present / Absent): Siyabonga Mkhize -Absent
4. Venue: Ritson DC1012 (Tutorial)
5. Minutes:- Schedule Recurring maintenance Use Case has been approved by the tutor by that it will be included in the system.
 - Mthobisi explained how Allocate student Use Case because he is the one who did coded it.
 - On the Allocate student Use Case students will be able scan a QR code in order to sign in and out of the residence through the system.
 - House Master will be able to review students that are allocated in their residences in order to approve them.
 - Students will be able to request swapping or change their residence that they are currently staying on to other residences.

6.Meeting closes; 11:00 am.